

AWS checklist and maintenance guide

Station name	KPL-L	
People present	MAV, RGP	instructed by: RST
Purpose	<u>standard maintenance</u> / installation / take-down / quick visit / emergency	
Date	26/7-2016	
Transport	<u>helicopter</u> / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	<u>warm</u> / cold / very cold / no wind / <u>some wind</u> / rain / cloud cover %: 0%	

Notes:

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	6 min	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)		☒ Y / N
Download data to PC and/or change CF Card (wait for green light)		Y / N
Name of logger program		
Photo or screen dump of values in fast scan mode		Y / N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		☒ Y / N
Mast tilt in boom direction (+ if radiometer tilting down)		°
Mast tilt across boom looking from radiometer (- if clockwise)		°
Boom direction relative to north	25	° true/magn = CW!
Box on wind sensor exactly towards mast? If not, measure direction.		☒ Y / N
Radiometer aligned with mast and inclinometer? If not, measure.		☒ Y / N
Temperature / humidity height (bottom of casing – surface)	278	cm
Sonic ranger on AWS height (membrane – surface)	288	cm
Wind sensor height (center of propeller – surface)	328	cm
Sonic ranger on stakes height (membrane – surface)	518	cm
Free length of stake – outer, holding boom with sonic ranger	—	cm
Free length of stake – middle	—	cm
Free length of stake – outer, holding boom without sonic ranger	—	cm
Free length of ablation hose on/above the ice surface	17.40	cm
Alt: depth of ablation hose using markings on hose		cm
Height of ablation hose fluid level above the ice surface	175	cm
Vertical difference in surface height (at station – at hose) (approx)	~55	cm
Length of thermistor string on surface from surface marking	778	cm
Height of surface irregularities	40	cm
Stake assembly still standing?		Y / ☒ N / more or less
Apparent damage to sensors? If yes, which?		Y / N

Notes:

Stakes all melted out.

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	120772	150710	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	00499	00958	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	Gears 29(9)	Gears 40	
Wind sensor	116618	144404	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly	1/2012		
T/hum assembly casing	60932242	61515230	- Fan spins and sounds OK: Y/N
T/hum assembly bracket	60881395	61531193	
Temperature probe in T/hum assembly	60946543	20067156	
HygroClip in T/hum assembly	60837428	8156816	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	4109	7155	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel	9B22B30		- Output OK: Y/N - Clean: Y/N
GPS antenna	2		- Stuck to top of logger enclosure: Y/N
Data logger	E4287	E10212	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	8392	12433	- LED green of flashing orange (active): Y/N
Multiplexer	E7531	E10797	- Clicking as it should: Y/N
Iridium modem IMEI number	2531510	4126980	- Transmissions tested: Y/N - Problems: Y/N
Barometer	4953185	6610222	
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string			- Old string left in ice: Y/N/cut
Ablation hose assembly			- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	Y/N
Name of logger program	2015.1
Photo or screen dump of values in fast scan mode	Y/N
Latitude (dd mm.mmm)	79°54.65333'
Longitude (dd mm.mmm)	021°04.97033'
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	-1 °
Mast tilt across boom looking from radiometer (- if clockwise)	-3 °
Boom direction relative to north (radiometer should point to true south)	0 ° true/magn
Box on wind sensor exactly towards mast? If not, change.	Y/N
Radiometer aligned with mast and inclinometer? If not, align.	Y/N
Temperature / humidity height (bottom of casing – surface)	same cm
Sonic ranger on AWS height (membrane – surface)	same cm
Wind sensor height (center of propeller – surface)	same cm
Sonic ranger on stakes height (membrane – surface)	109 cm
Free length of stake – outer, holding boom with sonic ranger	127 cm
Free length of stake – middle	121 cm
Free length of stake – outer, holding boom without sonic ranger	126 cm
Free length of ablation hose on/above the ice surface	— cm
Alt: depth of ablation hose using markings on hose	1395 cm
Height of ablation hose fluid level above the ice surface	180 cm
Vertical difference in surface height (at station – at hose) (approx)	140 cm
Length of thermistor string on surface from surface marking	= 4 cm
Height of surface irregularities	40 cm

Notes:

16 stakes left at site

AWS checklist and maintenance guide

Station name	KPL-0
People present	NAV, RSC, P.164 K852 instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency
Date	26/3-2016
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %:

Notes:

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC		+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)		Y / N
Download data to PC and/or change CF Card (wait for green light)		Y / N
Name of logger program		
Photo or screen dump of values in fast scan mode		Y / N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		Y / N
Mast tilt in boom direction (+ if radiometer tilting down)		°
Mast tilt across boom looking from radiometer (- if clockwise)		°
Boom direction relative to north		° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.		☒ Y / N
Radiometer aligned with mast and inclinometer? If not, measure.	256	Y / N
Temperature / humidity height (bottom of casing – surface)	236	cm
Sonic ranger on AWS height (membrane – surface)	232	cm
Wind sensor height (center of propeller – surface)	274	cm
Sonic ranger on stakes height (membrane – surface)	119	cm
Free length of stake – outer, holding boom with sonic ranger	175	cm
Free length of stake – middle	180	cm
Free length of stake – outer, holding boom without sonic ranger	173	cm
Free length of ablation hose on/above the ice surface	not visible	cm
Alt: depth of ablation hose using markings on hose	45	cm
Height of ablation hose fluid level above the ice surface	45	cm
Vertical difference in surface height (at station – at hose) (approx)		cm
Length of thermistor string on surface from surface marking		cm
Height of surface irregularities		cm
Stake assembly still standing?		Y / N / more or less
Apparent damage to sensors? If yes, which?		Y / N

Notes:

højde bærer 64 cm

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	120773	150711	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	01406	06888	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna		600541	
Wind sensor	116619	144398	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly			
T/hum assembly casing	60932241	61531191	- Fan spins and sounds OK: Y/N
T/hum assembly bracket	60786661	61515037	
Temperature probe in T/hum assembly	60886784	61578053	
HygroClip in T/hum assembly	60827600	20067158	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	4286		- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel	9B150413		- Output OK: Y/N - Clean: Y/N
GPS antenna			- Stuck to top of logger enclosure: Y/N
Data logger	E4288		- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	8393		- LED green of flashing orange (active): Y/N
Multiplexer	E7532		- Clicking as it should: Y/N
Iridium modem IMEI number	2930510	2901510	- Transmissions tested: Y/N - Problems: Y/N
Barometer	4953191		
Sonic ranger on stakes	4285	7157	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string	R34		- Old string left in ice: Y/N/cut
Ablation hose assembly			- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

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Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north (radiometer should point to true south)	° true/magn
Box on wind sensor exactly towards mast? If not, change.	Y / N
Radiometer aligned with mast and inclinometer? If not, align.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm

Notes: